Document 9: State Serialization & Session Persistence Specification

9.1 Host Automation State Integration

The plugin preserves the precise visual state of the drawing canvas and pedal routing matrix when a user saves a project within their Digital Audio Workstation (DAW). The system leverages the native host serialization wrappers (`getStateInformation` and `setStateInformation`) to manage data transfer.

9.2 Binary Payload Layout

The total session state is packed into a sequential, fixed-length binary large object (BLOB) memory stream containing four distinct segments:

- **Header Block (4 bytes):** Magic identification bytes and versioning tracking integers to guarantee backwards compatibility.
- **Canvas Matrix Block (16,384 bytes):** A direct, raw copy of the 16 KB uint8_t grid buffer.
- **Pedal Layout Configuration (6 bytes):** 6 sequential bytes mapping the active pedal types across Slot 0 through Slot 5 (using a numeric index representation where 0 equals unpopulated).
- **State Machine Flag (1 byte):** Explicit lock condition setting (0 for UNLOCKED, 1 for LOCKED).

Total serialized memory block size is fixed at exactly 16,395 bytes.

9.3 Deserialization and Reconstitution Pipeline

When the DAW restores a saved session, the payload stream is read sequentially to restore the plugin state:

1. The canvas memory block is copied directly back into the 128x128 GUI thread frame buffer.
2. The pedal configuration indices populate the active routing matrix structures.
3. The system executes an instantaneous, synchronous row allocation calculation using the logic in Document 2.
4. If any non-zero cells exist in the grid data, the state machine is forced into the LOCKED position, freezing the physical pedal assignments and making the canvas immediately ready for DSP processing.